



College of Basic and Applied Sciences
School of Engineering Sciences · Department of Computer Engineering

CPEN 208 — Project 1

Relational Database & Web Application
for the Computer Engineering Department

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Course: CPEN 208 — Introduction to Software Engineering

Semester: Second Semester, 2025/2026

Lecturer: Mr. Asiamah

PostgreSQL

Next.js 14

TypeScript

Tailwind CSS

GitHub: <https://github.com/doxaodoi/project-1-2>

1. Overview

This project delivers a relational database for a software the Computer Engineering Department intends to build, together with a Next.js 14 web application. Crucially, the five required items are implemented as **functionalities the user can perform** — not merely screens that display data. Each maps to a real action against the database:

#	Functionality	What the user can do
1	Student personal information	View their record and edit their contact details
2	Student fees payments	View the fee statement and make a payment (balance updates)
3	Course enrollment	Register for and drop current-semester courses
4	Lecturers to course assignment	View which lecturer teaches each course
5	Lecturers to TA assignment	View which TA assists each lecturer

A database function, `finance.get_outstanding_fees()`, computes every student's outstanding balance and returns it as a JSON array, as required.

1.1 Assumptions

- Database name `coe_dept`, split into four schemas: `people`, `academic`, `finance`, `auth`.
- Sample data is the real CPEN 208 class roster (70 students); emails follow the University pattern `<studentID>@st.ug.edu.gh`.
- Two academic terms are seeded so both grades and registration are meaningful: **Semester 1, 2025/2026** is completed (carries letter grades / results) and **Semester 2, 2025/2026** is the current term whose courses are open for registration.
- Each student is billed three items totalling **GH¢12,300.00**; payments vary to create realistic outstanding balances.
- Every student has a login account (`auth.users`) with the demo password `Student@123`.

2. Database Design

Four schemas hold the tables. Every relationship is enforced with a foreign key.

people

programs

program_id (PK), code, name, degree

students

student_id (PK), full_name, email, phone,
date_of_birth, level, program_id → programs

lecturers

lecturer_id (PK), staff_no, title, full_name, email,
academic_rank

teaching_assistants

ta_id (PK), full_name, email, student_id → students

academic

courses

course_id (PK), code, title, credit_hours, level,
semester

enrollments

enrollment_id (PK), student_id → students,
course_id → courses, academic_year, semester,
grade

lecturer_course_assignment

lecturer_id → lecturers, course_id → courses,
academic_year, semester

lecturer_ta_assignment

lecturer_id → lecturers, ta_id → teaching_assistants,
course_id → courses

finance

fee_bills

bill_id (PK), student_id → students, academic_year,
semester, description, amount_due

payments

payment_id (PK), student_id → students, amount,
paid_on, method, reference

auth

users

user_id (PK), student_id → students (unique), email
(unique), password_hash, role

2.1 Build scripts (run in pgAdmin, in order)

Script	Purpose
00_create_database.sql	Creates the <code>coe_dept</code> database
01_schemas.sql	Creates the four schemas
02_tables.sql	Creates all tables, keys and indexes
03_functions.sql	Creates the outstanding-fees functions
04_seed.sql	Inserts the 70-student roster and all sample data (both terms)

05_verify.sql	Row-count checks and a demonstration of the JSON function
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3. The Outstanding-Fees Function

The required function aggregates each student's total bills and total payments and returns `billed` – `paid` for every student, in a single JSON array.

```
CREATE OR REPLACE FUNCTION finance.get_outstanding_fees()
RETURNS JSON LANGUAGE sql STABLE AS $$
  SELECT COALESCE(json_agg(row_to_json(t) ORDER BY t.full_name), '[]'::json)
  FROM (
    SELECT s.student_id, s.full_name, s.email,
           COALESCE(b.total_billed, 0) AS total_billed,
           COALESCE(p.total_paid, 0) AS total_paid,
           COALESCE(b.total_billed, 0) - COALESCE(p.total_paid, 0) AS outstanding
    FROM people.students s
    LEFT JOIN (SELECT student_id, SUM(amount_due) AS total_billed
               FROM finance.fee_bills GROUP BY student_id) b ON b.student_id =
s.student_id
    LEFT JOIN (SELECT student_id, SUM(amount) AS total_paid
               FROM finance.payments GROUP BY student_id) p ON p.student_id =
s.student_id
    ) t;
$$;
```

Calling `SELECT finance.get_outstanding_fees();` returns a 70-element array:

```
[
  { "student_id": 22384451, "full_name": "Abu Neaquittae Golda",
    "total_billed": 12300.00, "total_paid": 12300.00, "outstanding": 0.00 },
  { "student_id": 22312110, "full_name": "David Kwame Odoi-Anim",
    "total_billed": 12300.00, "total_paid": 9000.00, "outstanding": 3300.00 }
  /* ... 70 students total ... */
]
```

A scalar helper, `finance.get_student_outstanding(student_id)`, returns one student's balance and is reused by the payments feature.

4. The Web Application — Functionalities in Action

The Next.js 14 application (App Router, TypeScript, Tailwind CSS) has login, register and a dashboard, and from the dashboard the user reaches each functionality. The app calls the Project 2 API; authentication uses a JWT stored in an HttpOnly cookie.

CoE Department Portal
Computer Engineering · University of Ghana

Sign in

Email
22312110@st.ug.edu.gh

Password
Student@123

Sign in

No account? [Register](#)

Demo: any seeded student e.g. 22312110@st.ug.edu.gh / Student@123

Login — seeded students sign in with their student-ID email and the demo password.

CoE Department Portal
Computer Engineering - University of Ghana - David Kwame Odoi-Anim

Sign out

DashboardPersonal InfoFeesRegistrationGradesAssignments

Signed in as
David Kwame Odoi-Anim
Student ID22312110
ProgrammeBSc Computer Engineering
Level200

Total billed
GH¢12,300.00

Total paid
GH¢9,000.00

Outstanding
GH¢3,300.00

WHAT WOULD YOU LIKE TO DO?



Personal Information
View and update your contact details.



Fees & Payments
See your statement and make a payment.



Course Registration
Register or drop courses this semester.



Grades / Results
Your results for completed courses.



Lecturers & TAs
Who teaches what, and TA assignments.

Dashboard — fee summary and a launch-pad to each of the five functionalities.

4.1 Functionality 1 — Personal Information (view & edit)

CoE Department Portal
Computer Engineering - University of Ghana - David Kwame Odoi-Anim

Sign out

DashboardPersonal InfoFeesRegistrationGradesAssignments

Personal Information

READ-ONLY DETAILS

Full name	David Kwame Odoi-Anim	Student ID	22312110
Programme	BSc Computer Engineering	Level	200
Date of birth	2004-04-26		

EDITABLE CONTACT DETAILS

Email

22312110@st.ug.edu.gh

Phone

+233241703923

Save changes

Read-only identity details plus an editable form for the student's contact email and phone.

4.2 Functionality 2 — Fees Payments (make a payment)

CoE Department Portal

Computer Engineering - University of Ghana - David Kwame Odoi-Anim

Sign out

Dashboard

Personal Info

Fees

Registration

Grades

Assignments

Fees & Payments

Total billed

GH¢12,300.00

Total paid

GH¢9,000.00

Outstanding

GH¢3,300.00

MAKE A PAYMENT

Amount (GHS)

up to 3300.00

Method

Bank

▼

Pay now

Payment History

Date	Amount	Method	Reference
2026-06-12	GH¢6,000.00	BANK	PAY-22312110-1
2026-07-14	GH¢3,000.00	MOMO	PAY-22312110-2

The statement (billed / paid / outstanding), a payment form, and payment history. A successful payment inserts a row and the outstanding balance drops immediately.

4.3 Functionality 3 — Course Registration (register & drop)

CoE Department Portal

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Sign out

Dashboard

Personal Info

Fees

Registration

Grades

Assignments

Course Registration

Second Semester, 2025/2026

Registered courses

2 course(s) · 6 credits

Code	Title	Credits	Lecturer	Action
CPEN 208	Introduction to Software Engineering	3	Prof. Elsie Adjei	Drop
CPEN 212	Digital Systems Design	3	Dr. Efua Danso	Drop

Available courses

Code	Title	Credits	Lecturer	Action
CBAS 210	Academic Writing II	3	Dr. Ama Boateng	Register
CPEN 214	Signals and Systems	3	Dr. Yaw Owusu	Register
CPEN 216	Electronic Circuits	3	Dr. Kwame Mensah	Register
MATH 224	Linear Algebra	3	Mr. Kofi Antwi	Register

Registered courses (with **Drop**) and available current-semester courses (with **Register**). Each button writes to `academic.enrollments`.

4.4 Grades / Results

Grades / Results

Completed courses

Indicative GPA

3.08

Code	Title	Credits	Year	Sem	Grade
CBAS 208	Academic Writing I	3	2025/2026	1	C
CPEN 202	Object Oriented Programming	3	2025/2026	1	B
CPEN 204	Data Structures and Algorithms	3	2025/2026	1	A
CPEN 206	Linear Circuits	3	2025/2026	1	A
CPEN 210	Digital Logic Design	3	2025/2026	1	B
MATH 221	Calculus II	3	2025/2026	1	C+

GPA is indicative, computed from published grades on a 4.0 scale.

Results for completed (Semester 1) courses, with an indicative GPA.

4.5 Functionalities 4 & 5 — Lecturer & TA assignments

CoE Department Portal

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Sign out

Dashboard

Personal Info

Fees

Registration

Grades

Assignments

Lecturers & Teaching Assistants

Lecturer → Course assignments

Course	Title	Lecturer	Year	Sem
CBAS 208	Academic Writing I	Prof. Elsie Adjei	2025/2026	1
CPEN 202	Object Oriented Programming	Dr. Kwame Mensah	2025/2026	1
CPEN 204	Data Structures and Algorithms	Dr. Ama Boateng	2025/2026	1
CPEN 206	Linear Circuits	Dr. Yaw Owusu	2025/2026	1
CPEN 210	Digital Logic Design	Dr. Efua Danso	2025/2026	1
MATH 221	Calculus II	Mr. Kofi Antwi	2025/2026	1
CBAS 210	Academic Writing II	Dr. Ama Boateng	2025/2026	2
CPEN 208	Introduction to Software Engineering	Prof. Elsie Adjei	2025/2026	2
CPEN 212	Digital Systems Design	Dr. Efua Danso	2025/2026	2
CPEN 214	Signals and Systems	Dr. Yaw Owusu	2025/2026	2
CPEN 216	Electronic Circuits	Dr. Kwame Mensah	2025/2026	2
MATH 224	Linear Algebra	Mr. Kofi Antwi	2025/2026	2

Lecturer → Teaching Assistant assignments

Lecturer	Teaching Assistant	Course	Year
Dr. Ama Boateng	Selorm Agbeko	CPEN 204	2025/2026
Dr. Efua Danso	Linda Ofori	CPEN 212	2025/2026
Prof. Elsie Adjei	Akosua Frimpong	CPEN 208	2025/2026
Prof. Elsie Adjei	Nana Kwabena Asare	CPEN 208	2025/2026

Read-only listings: which lecturer teaches each course, and which TA assists each lecturer.

5. How to Run

1. **Database:** in pgAdmin run `00_create_database.sql` on the `postgres` database, then `01 – 05` on `coe_dept` (or restore `coe_dept_backup.backup`).
2. **API (Project 2):** `cd "PROJECT 2/api"`, set `DB_PASSWORD`, then `mvn spring-boot:run` (port 8080).
3. **Front end:** `cd "PROJECT 1/web"`, copy `.env.example` to `.env.local`, then `npm install && npm run dev` and open `http://localhost:3000`.
4. Sign in with `22312110@st.ug.edu.gh` / `Student@123` and try each functionality.

Deliverables in this folder: SQL scripts and backup in `PROJECT 1/database` ; front-end source in `PROJECT 1/web` ; full source on GitHub at <https://github.com/doxaodoi/project-1-2> .